CS 70 离散数学 和 概率论
笔记 13
春季 2026 Course Notes
Introduction to 离散的 概率
概率论 hasits origins in gambling—analyzing cardgames, dice, roulette wheels. Todayit is an essentialtoolinengineeringandthesciences. NolesssoinEECS,whereitsuseiswidespreadinalgorithms, systems,control,signalprocessing,learningtheory,andAI.

Herearesometypicalstatementsthatyoumightseeconcerningprobability:
1. Thechanceofgettingaflushina5-cardpokerhandisabout2in1000.
2. The chance that this randomized primality testing 算法 outputs “prime” when the input is 非 primeisatmostoneinatrillion.
3. In this load-balancing scheme, the 概率 that any processor has to deal with more than 12 re- questsisnegligible.
4. Theaveragetimebetweensystemfailuresisabout3days.
5. Thereisa30%chanceofamagnitude8.0earthquakeinNorthernCaliforniabefore2040.

Implicitinallsuchstatementsisthenotionofanunderlyingprobabilityspace. Thismaybetheresultofa

randomexperimentthatwehaveourselvesconstructed(asin1,2and3above),orsomemodelwebuildof therealworld(asin4and5above). Noneofthesestatementsmakessenseunlesswespecifytheprobability spacewearetalkingabout: forthisreason,statementslike5(whicharetypicallymadewithoutthiscontext) arealmostcontent-free.
In this 笔记, we will try to understand all this more clearly. The first important notion here is that of a random experiment. We will start by introducing the space of all possible outcomes of the experiment, calledasamplespace. Eachelementofthesamplespaceisassignedaprobabilitywhichtellsushowlikely theoutcomeistooccurwhenweactuallyperformtheexperiment.

1 Random Experiments

Typically, a random experiment consists of drawing a sample of k elements from a 集合 S of cardinality n. The possible outcomes of such an experiment are exactly the objects that we counted in 笔记 12. Recall

fromNote12thatweconsideredfourpossiblescenariosforcounting,dependinguponwhetherwesampled with 或 without replacement, 和 whether the order in 其中 the k elements are chosen does 或 does 非 matter. The same will be the case 对于 our random experiments. The 结果 of a random experiment is called asample point, andthe 样本空间(often denotedby) isthe 集合 ofall possible outcomesof the experiment.
An 例子 of such an experiment is tossing a coin 4 times. In this case, S={H,T} 和 we are drawing 4 elements with replacement. HTHT is an 例子 of a sample point 和 the 样本空间 has 16 elements,asillustratedinthefollowingpicture:

HHHH HTHH THHH TTHH
HHHT HTHT THHT TTHT
=
HHTH HTTH THTH TTTH
HHTT HTTT THTT TTTT

Howdowedeterminethechanceofeachparticularoutcome,suchasHHTH,ofourexperiment? Inorder

todothis,weneedtodefinetheprobabilityforeachsamplepoint,asdescribedbelow.

2 概率 Spaces

Aprobabilityspaceisasamplespace ,
togetherwithaprobabilityP[ω](oftenalsodenotedasPr[ω])对于
eachsamplepointω,suchthat

• (Non-negativity): 0≤P[ω]≤1forallω ∈ .
• (Sumto1): ∑
P[ω]=1,i.e.,thesumoftheprobabilitiesoveralloutcomesis1.
ω∈

(Technically, P is a 实数-valued 函数 on a certain collection of subsets of , 和 is called 概率 measure.) The easiest way to assign probabilities to sample points is to do it uniformly: 如果 | |=N, 那么
P[ω]= 1, ω ∈ . 对于 例子, 如果 we toss a fair coin 4 times, each of the 16 sample points (as pictured
N
above)isassignedprobability 1 .
Wewillseeexamplesofnon-uniformprobabilityspacessoon.
16

After performing an experiment, we are often interested in knowing whether a certain 事件 occurred. 对于 例子, we might be interested in the 事件 that there were exactly 2 H’s in four tosses of the coin. How do we formally 定义 the concept of an 事件 in terms of the 样本空间 ? The 答案 is to identify the 事件 “exactly 2 H’s” with the 集合 of all those outcomes in 其中 存在 exactly two H’s, 即:
{HHTT,HTHT,HTTH,THHT,THTH,TTHH} . 因此, formally an 事件 A is just a 子集 of the samplespace ,i.e.,A .

How should we 定义 the 概率 of an 事件 A? Naturally, we should just add up the probabilities of thesamplepointsinA.

ForanyeventA ,wedefinetheprobabilityofAtobe
P[A]= ∑ P[ω].
ω∈A

Notethat0≤P[A]≤1forallA ,andP[ ]=1.

TheprobabilityofgettingexactlytwoH’sinfourcoin tossescanbecalculatedusingthisdefinitionasfollows. EventAconsistsofallsequencesthathaveexactly twoH’s,andso|A|= (cid:0)4(cid:1) =6.
Forthisexample,thereare24=16possibleoutcomesforflippingfourcoins.
2
因此, each sample point ω ∈A has 概率 1 ; 和 因为 存在 six sample points in A, 我们得到
16
P[A]=6• 1 = 3.
16 8

3 Examples

We will now look at examples of random experiments 和 their corresponding sample spaces, along with possibleprobabilityspacesandevents.

3.1 Coin Tosses
SupposewehaveacoinwithP[H]=
pandP[T]=1 p,andourexperimentconsistsofflippingthecoin4
times.
Thesamplespace consistsofthesixteenpossiblesequencesofH’sandT’sshowninthefigureon
thepreviouspage.
The 概率 space depends on p. 如果 p= 1 the probabilities are
assigned uniformly; the 概率 of
2
each sample point is 1 . What 如果 p= 2? 那么 the probabilities
are different. 对于 例子, P[HHHH]=
16 3
2×2×2×2 = 16,whileP[TTHH]= 1×1×2×2 = 4 . (笔记:
Wehavesimplymultipliedprobabilities
3 3 3 3 81 3 3 3 3 81
here; we will explain later why this is the correct thing to
do 对于 this 例子. It is 非 always OK to
multiplyprobabilitieslikethis.)
What type of events can we 考虑 in this setting? 令 A be the 事件
that all four coin tosses are the
same. 那么 A={HHHH,TTTT}. HHHH has 概率
(cid:0)2(cid:1)4
和 TTTT has 概率
(cid:0)1(cid:1)4
. 因此,
3 3
P[A]=P[HHHH]+P[TTTT]= (cid:0)2(cid:1)4 + (cid:0)1(cid:1)4 =
17.
3 3 81
Next,considertheeventBthatthereareexactlytwoheads.
Theprobabilityofanyparticularoutcomewith
two heads (such as HTHT) is
(cid:0)2(cid:1)2(cid:0)1(cid:1)2
. How many such outcomes are there? 存在
(cid:0)4(cid:1)
=6 ways
3 3 2
of choosing the positions of the heads, 和 these choices
completely specify the 序列. 所以, P[B] =
6 (cid:0)2(cid:1)2(cid:0)1(cid:1)2 = 24 = 8 .
3 3 81 27

Moregenerally,ifweflipthecoinntimes,wegetasamplespace ofcardinality2n.
Thesamplepointsare
all possible length-n sequences of H’s 和 T’s. 如果 the coin
has P[H]= p, 和 如果 we 考虑 any 序列

ofncoinflipswithexactlyrH’s,thentheprobabilityofthissequenceis
pr(1 p)n r.
Now 考虑 the eventC that we get exactly r H’s when we flip the
coin n times. This 事件 consists of
exactly (cid:0)n(cid:1) samplepointsandeachhasprobability
pr(1 p)n r. 所以,P[C]= (cid:0)n(cid:1) pr(1 p)n r.
r r
Biasedcointossingsequencesshowupinmanycontexts:
forexample,theymightmodelthebehaviorofn

independenttrialsofafaultysystem,whichfailseachtimewithprobability
p.

3.2 Rolling Dice
考虑 rolling two fair dice. In this experiment, = {(i,j) : 1
⩽ i,j ⩽ 6}. The 概率 space is
均匀的,i.e.,allsamplepointshavethesameprobability 1 = 1 .
因此,theprobabilityofanyeventAis
| | 36
#ofsamplepointsinA |A|
P[A]= = .
#ofsamplepointsin | |

所以,foruniformspaces,computingprobabilitiesreducestocountingsamplepoints!
Now 考虑 two events: the 事件 A that the sum of the dice is at
least 10 和 the 事件 B that 存在 at
least one 6. By enumerating the sample points contained in
each 事件, it can be easily shown that |A|=6
和|B|=11. 那么,bytheobservationabove,itfollowsthatP[A]= 6 = 1
andP[B]= 11.
36 6 36
3.3 Card Shuffling

Considertherandomexperimentofshufflingadeckofstandardplayingcards.
Here, isequaltothesetof
all52!permutationsofthedeck.
Weassumethattheprobabilityspaceisuniform.
(Notethatwearereally
talking about an idealized mathematical model of shuffling
here; in 实数 life, there will always be a bit of
bias in our shuffling. 然而, the mathematical model is close
enough to be useful.) We can generalize

thisexperimenttorandomly“shuffling”nitems,forarbitraryn.
(Forcardshuffling,n=52.) Forexample,

supposewecollectthehomeworksofnstudentsinaclass,thenredistributethemrandomly,oneperstudent.

Theresultisarandompermutationofthehomeworks,witheachoutcomehavingprobability
1.
n!
3.4 Poker Hands
Here’sanotherexperiment:
shufflingadeckofcardsanddealingapokerhand.
Inthiscase,Sisthesetof52
cards 和 our 样本空间 ={allpossiblepokerhands}, 其中 corresponds
to choosing k=5 objects
without replacement from a 集合 of size n=52 其中 order does 非
matter. 因此, as we saw in 笔记 12,
| |= (cid:0)52(cid:1) = 52×51×50×49×48 =2,598,960.
Assumingthatthedeckiswellshuffled,theprobabilityofeach
5 5×4×3×2×1

outcomeisequallylikelyandwearethereforedealingwithauniformprobabilityspace.
LetAbetheeventthatthepokerhandisaflush.
(Forthosewhoarenotfamiliarwithpoker,aflushisahand
in 其中 all cards have the same suit, say Hearts.) 因为 the 概率
space is 均匀的, computing P[A]
reduces to simply computing |A|, the number of poker hands
that are flushes. 存在 13 cards in each
suit,sothenumberofflushesineachsuitis
(cid:0)13(cid:1)

. Thetotalnumberofflushesistherefore4×
(cid:0)13(cid:1)
. Thenwe
5 5
have
4×
(cid:0)13(cid:1)
13! 5!47! 13・12・11・10・9
P[handisaflush]= 5 =4× × =4× ≈0.002.
(cid:0)52(cid:1) 5!8! 52! 52・51・50・49・48
5
3.5 Balls 和 Bins

Inthisexperiment,wewillthrow20(labeled)ballsinto10(labeled)bins.
Assumethateachballisequally
likelytolandinanybin,regardlessofwhathappenstotheotherballs.
如果 you wish to understand this situation in terms of
sampling a 序列 of k elements from a 集合 S of
cardinality n: the 集合 S consists of the 10 bins, 和 we are
sampling with replacement k=20 times. The
orderofsamplingmatters,sincetheballsarelabeled.
The 样本空间 is equal to {(b ,b ,...,b):1≤b ≤10
foreachi=1,...,20}, 其中 the compo-
1 2 20 i
nentb denotesthebininwhichballilands.
Thecardinality| |ofthesamplespaceisequalto1020,因为
i
each 元素 b in the 序列 has 10 possible choices 和 存在 20 elements
in the 序列. More
i
generally, 如果 we throw m balls into n bins, 我们有 a 样本空间 of
size nm. The 概率 space is
均匀的;aswesaidearlier,eachballisequallylikelytolandinanybin.
令 A be the 事件 that bin 1 is empty. 因为 the 概率 space is 均匀的,
we simply need to count
how many outcomes have this property. This is exactly the
number of ways all 20 balls can 秋季 into the
remainingninebins,whichis920. 因此,P[A]= 920 = (cid:0) 9
(cid:1)20 ≈0.12.
1020 10
LetBbetheeventthatbin1containsatleastoneball.
Thiseventisthecomplement AofA,i.e.,itconsists
ofpreciselythosesamplepointswhicharenotinA.
SoP[B]=1 P[A]≈0.88. Moregenerally,ifwethrow
mballsintonbins,wehave:
(cid:18)
n 1
(cid:19)m (cid:18)
1
(cid:19)m
P[bin1isempty]= = 1 .
n n
As we shall see, balls 和 bins is another 概率 space that shows
up very often in Computer Science:

forexample,wecanthinkofitasmodelingaloadbalancingscheme,inwhicheachjobissenttoarandom
processor.

It is also a more general model 对于 problems 我们有 previously
considered. 对于 例子, flipping a fair

coin4timesisthespecialcaseinwhichthenumberofballs(m)is4andthenumberofbins(n)is2.
Rolling
twodiceisthespecialcaseinwhichm=2andn=6.
3.6 Birthday Paradox

The“birthdayparadox”isaremarkablephenomenonthatexaminesthechancesthattwopeopleinagroup
havethesamebirthday.
Itisa“paradox”notbecauseofalogicalcontradiction,butbecauseitgoesagainst
intuition. 对于 ease of calculation, we take the number of
days in a year to be 365. 那么 S={1,...,365},

andtherandomexperimentconsistsofdrawingasampleofnelementsfromSwithreplacement,wherethe
elementsarethebirthdatesofnpeopleinagroup. 那么| |=365n.
Thisisbecauseeachsamplepointis
asequenceofpossiblebirthdaysfornpeople;
sotherearenpointsinthesequenceandeachpointhas365
possiblevalues.
令 A be the 事件 that at least a pair of people have the same
birthday. 如果 we want to determine P[A], it

mightbesimplertoinsteadcomputetheprobabilityofthecomplementofA;i.e.,P[A],whereAistheevent
thatnotwopeoplehavethesamebirthday.
SinceP[A]=1 P[A],wecantheneasilycomputeP[A].

Weareagainworkinginauniformprobabilityspace,sowejustneedtodetermine|A|.
Equivalently,weare

computingthenumberofwaysfornotwopeopletohavethesamebirthday.
Thereare365choicesforthe

firstperson,364forthesecond,...,365 n+1choicesforthenthperson,foratotalof365×364×・・・×
(365 n+1). This is simply an application of the First Rule
of Counting from 笔记 12; we are sampling

withoutreplacement(sinceweneedtoavoidcollisions)andtheordermatters.
Insummary,wehave
|A| 365×364×・・・×(365 n+1)
P[A]= = ,
| | 365n
soP[A]=1 P[A]=1 365×364×・・・×(365 n+1) .
ThisallowsustocomputeP[A]asafunctionofthenumbern
365n
ofpeople. Ofcourse,asnincreasesP[A]increases.
Infact,withn=23people,youshouldbewillingtobet

thatatleastapairofpeoplehavethesamebirthday,sincethenP[A]islargerthan50%.
Forn=60people,
P[A]isover99%.
3.7 The Monty Hall 题目

Thisexampleislooselybasedonan(in)famous1970sTVgameshowhostedbyMontyHall.
Acontestant

isshownthreedoors;behindoneofthedoorsisavaluableprize(acar),andbehindtheothertwoaregoats.
Thecontestantpicksadoor(butdoesnotopenit).
ThenHall’sassistant(Carol)opensoneoftheothertwo

doors,revealingagoat(sinceCarolknowswheretheprizeis,shecanalwaysdothis). Thecontestantisthen

giventheoptionofstickingwithhis/hercurrentdoor,orswitchingtotheotherunopenedone. Thecontestant winstheprizeifandonlyiftheirchosendooristhecorrectone. Thequestionis: Doesthecontestanthave abetterchanceofwinningifhe/sheswitchesdoors?

Intuitively,itseemsobviousthatsincethereareonlytworemainingdoorsafterCarolopensone,theymust haveequalprobability. Soyoumaybetemptedtojumptotheconclusionthatitshouldnotmatterwhetheror notthecontestantstaysorswitches. Wewillseethatactually,thecontestanthasabetterchanceofpicking the car 如果 he 或 she uses the switching strategy. We will first give an intuitive pictorial argument, 和 那么 takeamorerigorousprobabilityapproachtotheproblem. To see why it is in the contestant’s best interests to switch, 考虑 the following. Initially when the contestant chooses the door, he 或 she has a 1 chance of picking the car. This must mean that the other
3

doors combined have a 2 chance of winning. 但是 after Carol opens a door with a goat behind it, how do
3
theprobabilitieschange? Well,thedoorthecontestantoriginallychosestillhasa 1 chanceofwinning,和
3
thedoorthatCarolopenedhasnochanceofwinning. Whataboutthelastdoor? Itmusthavea 2 chanceof
3
containing the car, 和 所以 the contestant has a higher chance of winning 如果 he 或 she switches doors. This argumentcanbesummedupnicelyinthefollowingpicture:
What is the 样本空间 here? Well, we can describe the 结果 of the game (up to the point 其中 the

contestantmakeshis/herfinaldecision)usingatripleoftheform(i,j,k),wherei,j,k∈{1,2,3}. Thevalues i,j,k respectively specify the location of the prize, the initial door chosen by the contestant, 和 the door openedbyCarol. Notethatsometriplesarenotpossible: e.g.,(1,2,1)isnot,becauseCarolneveropensthe prize door. Thinking of the 样本空间 as a 树 structure, in 其中 first i is chosen, 那么 j, 和 finally k (dependingoniand j),weseethattherearexactly12samplepoints.

Assigningprobabilitiestothesamplepointshererequirespinningdownsomeassumptions:

- Theprizeisequallylikelytobebehindanyofthethreedoors.
- Initially,thecontestantisequallylikelytopickanyofthethreedoors.
- 如果 the contestant happens to pick the prize door (所以 存在 two possible doors 对于 Carol to open), Carolisequallylikelytopickeitherone. (Actuallyyourcalculationwillhavethesameresultnomatter howCarolpicksthedoor.)

Fromthis,wecanassignaprobabilitytoeverysamplepoint.
Forexample,thepoint(2,1,3)correspondsto
theprizebeingplacedbehinddoor2(withprobability
$\frac{1}{3}$),thecontestantpickingdoor1(withprobability $\frac{1}{3}$),

andCarolopeningdoor3(withprobability1,becauseshehasnochoice).
所以

$$P[(2,1,3)] = \frac{1}{3}\times\frac{1}{3}\times 1 = \frac{1}{9}.$$

[笔记: Again we are multiplying probabilities here, without
proper justification!] 笔记 that 存在 six
outcomesofthistype,characterizedbyhaving$i \neq j$(andhencekmustbedifferentfromboth). Ontheother
hand,wehave

$$P[(1,1,2)] = \frac{1}{3}\times\frac{1}{3}\times\frac{1}{2} = \frac{1}{18}.$$

Andtherearesixoutcomesofthistype,having$i = j$.
Thesearetheonlypossibleoutcomes,sowehavecom-
pletelydefinedourprobabilityspace.
Justtocheckourarithmetic,wenotethatthesumoftheprobabilities
ofalloutcomesis$(6\times\frac{1}{9})+(6\times\frac{1}{18})=1$.

令’sreturntotheMontyHallproblem.
Recallthatwewanttoinvestigatetherelativemeritsofthe“sticking”
strategy 和 the “switching” strategy. 令’s 假设 the contestant
decides to switch doors. The eventW
weareinterestedinistheeventthatthecontestantwins.
Whichsamplepoints(i,j,k)areinW? Well,因为

thecontestantisswitchingdoors,theirinitialchoice j
cannotbeequaltotheprizedoor,whichisi. Andall

outcomesofthistypecorrespondtoawinforthecontestant,becauseCarolmustopenthesecondnon-prize
door, leaving the contestant to switch to the prize door.
SoW consists of all outcomes of the first type in
our earlier analysis; 回忆 存在 six of these, each with 概率 $\frac{1}{9}$. 所以
$P[W] = \frac{6}{9} = \frac{2}{3}$. 即,

usingtheswitchingstrategy,thecontestantwinswithprobability
$\frac{2}{3}$. Itshouldbeintuitivelyclear(andeasy
to check formally—try it!) that under the sticking strategy
their 概率 of winning is $\frac{1}{3}$. (In this case,
thecontestantisreallyjustpickingasinglerandomdoor.)
Sobyswitching,thecontestantactuallyimproves
theiroddsbyahugeamount!

4 Summary

The Monty Hall 例子 well illustrates the importance of doing
概率 calculations systematically,
ratherthan“intuitively.”
Recallthekeystepsinallourcalculations:

- Whatisthesamplespace(i.e.,theexperimentanditssetofpossibleoutcomes)?
- Whatistheprobabilityofeachoutcome(samplepoint)?
- Whatistheeventweareinterestedin(i.e.,whichsubsetofthesamplespace)?

• Finally, 计算 the 概率 of the 事件 by adding up the probabilities of the sample points containedinit.

Whenever you meet a 概率 题目, you should always go back to these basics to avoid potential pitfalls. Evenexperiencedresearchersmakemistakeswhentheyforgettodothis—witnessmanyerroneous

“proofs”,submittedbypeoplewithPhDstonewspapersatthetime,oftheclaimthattheswitchingstrategy intheMontyHallproblemdoesnotimprovetheodds.